ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C10 · ENERGY RESOURCES

Renewable or not?

★☆☆
Grade 5

Learning goals

- Classify energy resources.
- Read data from a table.

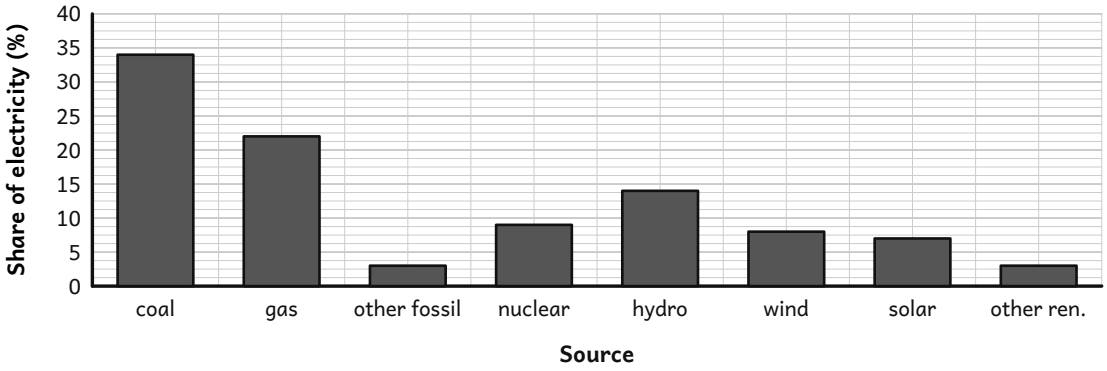
1 **Sort the energy resources.** [8]

SORT THESE

natural gas **biogas** **wind** **solar** **uranium** **hydro** **oil** **coal**

Renewable	Non-renewable
solar	coal
wind	oil
hydro	natural gas
biogas	uranium

2 **Draw a bar chart of the world's electricity in 2024.** [4]



3 **Use the chart.** [2]

a) Which single source made the most electricity?

coal (34%)

b) Add up wind + solar + hydro + other renewables.

8 + 7 + 14 + 3 = 32%

ANSWER KEY

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THINK IT THROUGH · C10 · ENERGY RESOURCES

Weighing up energy choices

★★☆
Grade 5

Learning goals

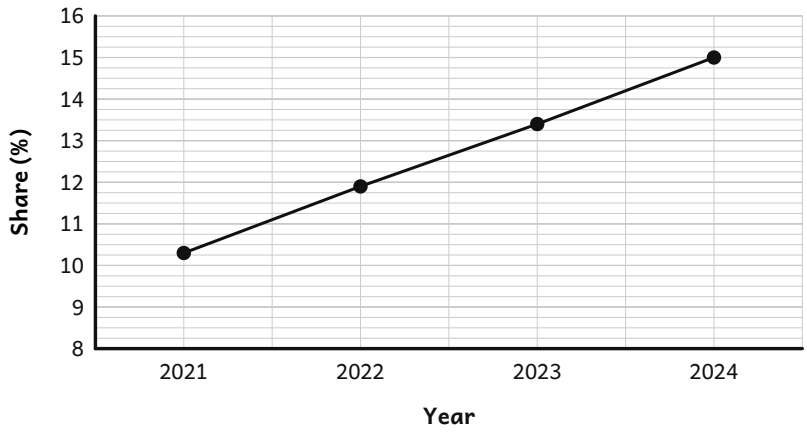
- Compare advantages and disadvantages.
- Interpret a trend.

1 **Complete the table.** [8]

Resource	One advantage	One disadvantage
coal	cheap and gives power all the time	releases lots of CO_2 and smoke; will run out
solar	free, clean sunlight	no power at night; needs batteries
wind	clean and renewable	only works when it is windy
hydro	clean and reliable	dams flood land and move people

2 **The share of the world's electricity from wind and solar together is shown below. Plot it as a line graph.** [3]

Year	2021	2022	2023	2024
Wind + solar (%)	10.3	11.9	13.4	15.0



3 **Describe the trend. Predict the share in 2025 if it continues.** [2]

It rises by about 1.5 percentage points each year; about 16.5% in 2025.

ANSWER KEY

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INVESTIGATE · C10 · ENERGY RESOURCES

Does the number of blades change a wind turbine's speed?

★★☆ Grade 5

Learning goals

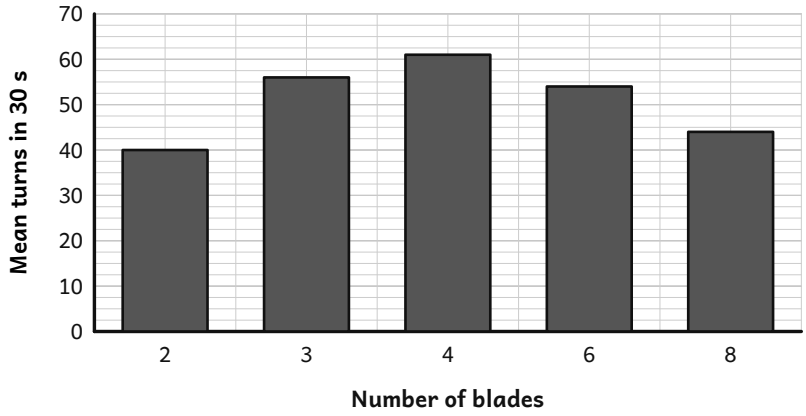
- Plan a fair test with a model.
- Analyse results.

Class 5A made paper wind turbines with 2, 3, 4, 6 and 8 blades. They put each one 1 m in front of the same fan and counted the turns in 30 seconds (three tries each).

1 **Complete the table: work out the mean (average) of the three tries.** [5]

Blades	Try 1	Try 2	Try 3	Mean turns
2	38	40	42	40
3	54	57	57	56
4	62	60	61	61
6	52	55	55	54
8	45	43	44	44

2 **Draw a bar chart of the mean turns.** [3]



3 **Describe the pattern. Which number of blades worked best?** [2]

Speed rose up to 4 blades, then fell. 4 blades worked best: more blades catch more wind, but too many add weight and block the air.

4 **Why did they do three tries and find the mean?** [1]

To make the results more reliable and reduce the effect of odd results.

Data: example results for practice.

ANSWER KEY

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REAL-WORLD CASE STUDY · C10 · ENERGY RESOURCES

Bangladesh's energy future

★★★
Grade 5

Learning goals

- Use data to evaluate a big decision.
- Consider different viewpoints.

Science | Geography | Economics | Citizenship

Bangladesh needs much more electricity for its homes, factories and growing cities. In 2023 it made over 85% of its electricity from fossil fuels: mostly natural gas, plus oil and a fast-growing amount of coal. Its own gas fields are running down, so it must buy more fuel from other countries, which is expensive.

At the same time, Bangladesh is one of the countries hit hardest by climate change. Many people say the country should build much more **solar** (on rooftops, over ponds and on unused land) and **wind** power along the coast, while saving energy. Others worry that solar needs a lot of land in a crowded country and does not work at night without batteries.

Based on Power Grid Company of Bangladesh data (2023) and public debate about Bangladesh's energy plans.

1 **Why must Bangladesh buy more fuel from other countries?** [2]

Its own natural gas fields are running down, but it needs more electricity.

2 **Complete the viewpoints.** [3]

Person	What might they want? Why?
a coastal farmer	clean energy, because climate change brings cyclones and salty water to their fields
a factory owner	reliable, cheap power all day so the factory never stops
a city child with asthma	less coal and oil burning, so the air is cleaner

3 **Write a short plan (3 actions) for green energy in Bangladesh. Use evidence.** [4]

e.g. rooftop solar on schools and factories; wind farms on the coast; LED lights and efficient fans to save energy; use gas, not coal, while renewables grow.

ANSWER KEY

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REVIEW AND CHALLENGE · C10 · ENERGY RESOURCES

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the words.**

F	L	G	B	S	W	Y	G	S	B
F	S	O	R	I	P	K	P	C	D
T	M	O	T	E	O	W	W	L	W
R	B	N	L	C	E	G	N	H	I
D	W	M	M	A	O	N	A	F	N
I	C	E	I	D	R	A	I	S	D
C	L	I	M	A	T	E	L	J	B
G	L	V	T	C	U	M	L	J	P
A	U	H	Y	D	R	O	F	M	N
S	P	R	S	F	I	O	V	D	K

biogas **climate** **coal** **gas** **green** **hydro** **solar** **wind**

2 **True or false?** [3]

Statement	True	False	If false, correct it
1. Natural gas is a renewable resource.		✔	It is a fossil fuel and will run out.
2. Most of Bangladesh's electricity in 2023 came from natural gas.	✔		
3. Nuclear power releases a lot of carbon dioxide.		✔	It releases almost none, but its fuel is non-renewable and its waste is dangerous.

3 **About what share of the world's electricity came from fossil fuels in 2024?** [1]

◯ about 20%
● **about 59%**
◯ about 90%

4 **Challenge. Your school wants to cut its electricity bill and its carbon footprint. Suggest three changes and explain each.** [3]

Solar panels on the roof (free, clean electricity); LED bulbs (use less energy); switch off fans and lights in empty rooms (no waste).